Volumes by the Disc and Washer Methods

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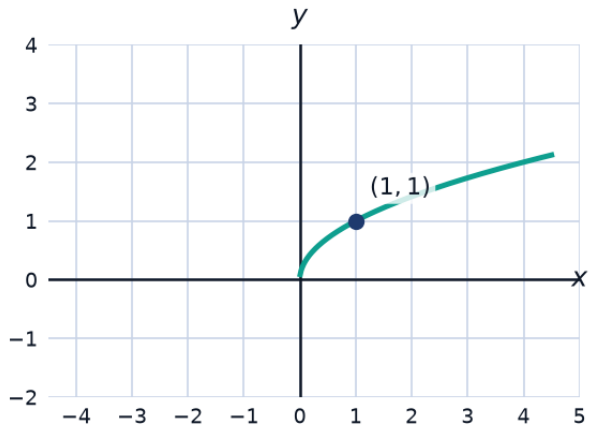
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Decompose the rational expression $\frac{5x^2 - 3x + 2}{(x-1)(x^2+1)}$ into partial fractions.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 2 · MediumFind the third-degree Taylor polynomial $P_3(x)$ for $f(x) = \sqrt{x}$ centered at $c = 4$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A student receives the following scaled marks on IB AA HL assessment: Paper 1: 85/110, Paper 2: 92/110, Paper 3: 44/55, and IA: 16/20. Calculate the student's overall weighted percentage.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Level 3 · Advanced / AP FRQLet $P_4(x)$ be the fourth-degree Maclaurin polynomial for $\cos(x)$. Use the Lagrange Error Bound to prove that \(|\cos(0.5) - P_4(0.5)|

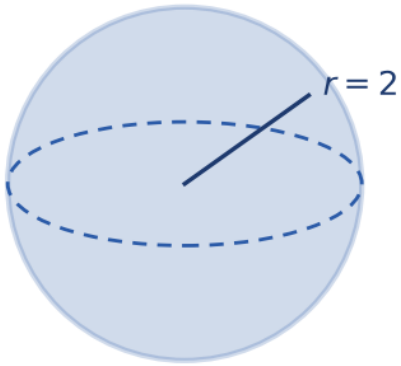
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 3 · Advanced / AP FRQThe alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^4}$ converges to sum S. If the sum is approximated by the fourth partial sum S_4, use the Alternating Series Error Bound to find an upper bound for the error $|S - S_4|$.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 1 · FoundationsFind the radius of convergence R for the power series $\sum_{n=0}^{\infty} n!\,(2x-1)^n$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 2 · Medium Find the third-degree Maclaurin polynomial $P_3(x)$ for $f(x) = \cos x$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the Maclaurin series for $\ln(1+2x)$ up to the x^3 term, and state its interval of convergence.

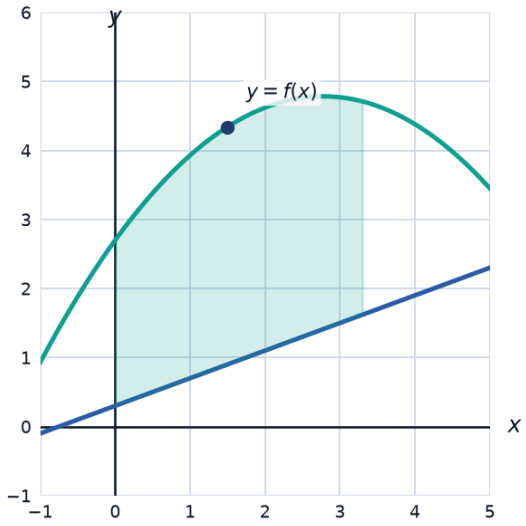
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 1 · FoundationsExpress the repeating decimal $0.272727\ldots = 0.\overline{27}$ as an infinite geometric series, and find its exact fraction representation in simplest form.

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10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Level 1 · Foundations Evaluate the indefinite integral $\int \frac{\cos(\ln x)}{x}\,dx$ for $x > 0$.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Applications of Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Applications of Derivatives · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Convergence Tests & Power Series · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Derivatives · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

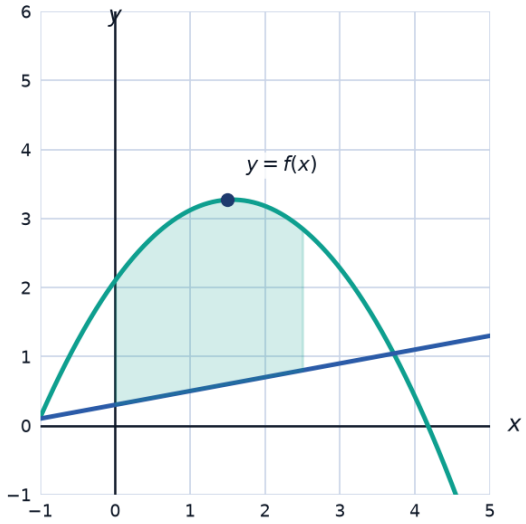
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Limits Graphically and Numerically

[Foundations & Precalculus Review · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

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16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

[Integrals · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 The Definition of the Derivative

[Parametric & Polar · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

[Partial Derivatives & the Gradient · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 The Chain Rule

[Sequences & Series · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

[Techniques of Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ about the x-axis.

a. 8π

b. 16π

c. 4π

d. $\frac{16\pi}{3}$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the volume when the region bounded by $y = x^2, x = 0,$ and $y = 4$ is revolved about the y-axis.

a. 8π

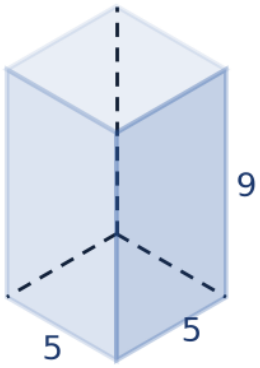
b. 16π

c. 4π

d. $\frac{32\pi}{5}$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the cross-sectional shape of a solid formed by the disc method?



a. A solid circular disk of area πR^2

b. A circular ring (washer)

c. A rectangle

d. A cylindrical shell

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The region enclosed by $y = x$ and $y = x^2$ is revolved about the x-axis. What is the resulting volume?

a. $\frac{2\pi}{15}$

b. $\frac{\pi}{6}$

c. $\frac{\pi}{15}$

d. $\frac{4\pi}{15}$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Write the integral for the volume when the region between $y = 2x$ and $y = x^2$ is revolved about the x-axis.

a. $\pi\int_0^2 (4x^2 - x^4)dx$

b. $\pi\int_0^2 (2x - x^2)^2 dx$

c. $\pi\int_0^2 (x^4 - 4x^2)dx$

d. $2\pi\int_0^2 (2x - x^2)dx$

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Why is $\pi\int (R - r)^2 dx$ INCORRECT for the washer method?

a. The area of a washer is $\pi R^2 - \pi r^2$, which is NOT equal to $\pi(R - r)^2$

b. It omits the factor of 2

c. It is only valid for shells

d. It produces negative volume

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The region bounded by $y = x^2$ and $y = 4$ is revolved about the line $y = 4$. What is the outer radius $R(x)$?

a. $R(x) = 4 - x^2$

b. $R(x) = x^2$

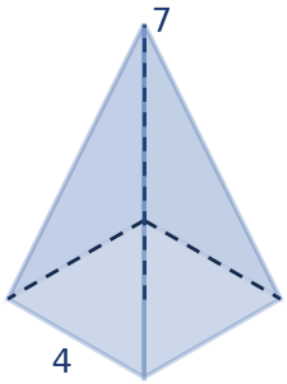
c. $R(x) = 4$

d. $R(x) = x^2 - 4$

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the volume in the above problem revolved about $y = 4$.



a. $\frac{256\pi}{15}$

b. $\frac{128\pi}{15}$

c. $\frac{64\pi}{5}$

d. 16π

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The region between $y = \sqrt{x}, y = 0,$ and $x = 4$ is revolved about the line $y = -1$. What are the washer radii?

a. $R(x) = \sqrt{x} + 1$ and $r(x) = 1$

b. $R(x) = \sqrt{x} - 1$ and $r(x) = -1$

c. $R(x) = \sqrt{x}$ and $r(x) = 1$

d. $R(x) = 4$ and $r(x) = \sqrt{x}$

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The base of a solid is the region enclosed by $y = 1 - x^2$ and the x-axis. Cross-sections perpendicular to the x-axis are squares. What is the volume?

a. $\frac{16}{15}$

b. $\frac{8}{15}$

c. $\frac{4}{3}$

d. $\frac{32}{15}$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If cross-sections perpendicular to the x-axis are semicircles with diameter on the base, what is the cross-sectional area formula?

a. $A(x) = \frac{\pi}{8}[s(x)]^2$

b. $A(x) = \frac{\pi}{2}[s(x)]^2$

c. $A(x) = \frac{\pi}{4}[s(x)]^2$

d. $A(x) = \pi[s(x)]^2$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If cross-sections are equilateral triangles with side length $s(x)$, what is the cross-sectional area formula?

a. $A(x) = \frac{\sqrt{3}}{4}[s(x)]^2$

b. $A(x) = \frac{1}{2}[s(x)]^2$

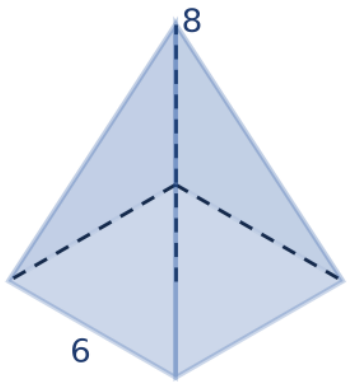
c. $A(x) = \frac{\sqrt{3}}{2}[s(x)]^2$

d. $A(x) = \frac{1}{4}[s(x)]^2$

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Volumes: for the same region rotated, which is larger: shell or washer?



a. Shell is always larger

b. They give the same volume

c. Washer is always larger

d. Depends on axis and region

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Washer method: volume when $R(x)$ is outer and $r(x)$ inner radius:

a. $\pi\int(R-r)^2dx$

b. $\pi\int(R^2-r^2)dx$

c. $2\pi\int(R-r)dx$

d. $\pi\int R^2dx-\pi\int r^2dx$ (same)

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Volume of cone: radius r, height h, using disc method:

a. $\frac{2}{3}\pi r^2h$

b. $\frac{1}{2}\pi r^2h$

c. πr^2h

d. $\frac{1}{3}\pi r^2h$

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The volume of the region $1 \le x^2 + y^2 + z^2 \le 4$ (shell between spheres of radii 1 and 2):

a. $4\pi/3$

b. 4π

c. 8π

d. $4\pi(8-1)/3 = 28\pi/3$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use spherical coordinates to find the volume of the first-octant part of the ball $x^2 + y^2 + z^2 \le 16$.

a. $\frac{256\pi}{3}$

b. $\frac{64\pi}{3}$

c. $\frac{32\pi}{3}$

d. $\frac{16\pi}{3}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Find the volume of the solid inside the sphere $\rho = 2$ and above the cone $\phi = \frac{\pi}{4}$.

a. $2\pi\left(\frac{8}{3} - \frac{4\sqrt{2}}{3}\right)$

b. $4\pi\left(\frac{8}{3} - \frac{4\sqrt{2}}{3}\right)$

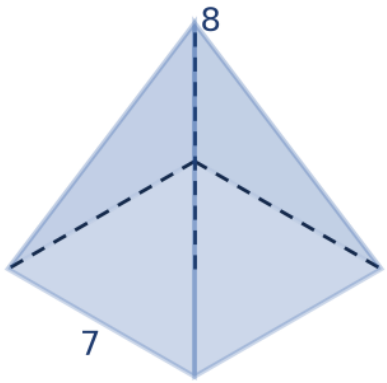
c. $\frac{16\pi}{3}$

d. $2\sqrt{2}\,\pi\left(\frac{8}{3} - \frac{4\sqrt{2}}{3}\right)$

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 The Chain Rule

The Jacobian of the transformation $x = r\cosh u, y = r\sinh u$ is r times:



a. $\cosh^2 u - \sinh^2 u = 1$

b. $\cosh u + \sinh u$

c. $\sinh^2 u$

d. r^2

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Centre of mass of lamina $D = [0,1] \times [0,1]$ with density $\delta = xy$:

a. $(1/2, 1/2)$

b. $(2/3, 2/3)$

c. $(3/4, 3/4)$

d. $(1, 1)$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Use polar coordinates to evaluate $\iint_R (x^2 + y^2)^3 \, dA$, where R is the disk $x^2 + y^2 \le 9$.

a. $\frac{6561}{4}$

b. $\frac{6561\pi}{4}$

c. $\frac{6561\pi}{2}$

d. $\frac{6561\pi}{8}$

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

The double integral over region D of $f = 1$ gives:

a. area of D

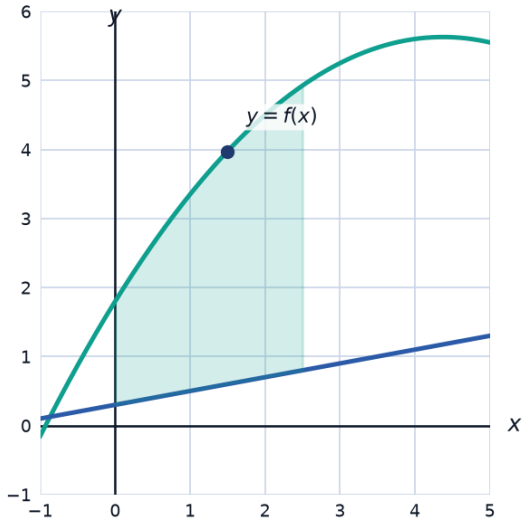
b. perimeter of D

c. volume

d. 0

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Triple integral $\iiint_V 1\, dV$ gives:



a. volume of V

b. 0

c. surface area

d. 1

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Use polar coordinates to evaluate $\iint_R \sqrt{x^2 + y^2}\, dA$, where R is the annulus $1 \le x^2 + y^2 \le 4$.

a. $\frac{14}{3}$

b. $\frac{14\pi}{3}$

c. $\frac{28\pi}{3}$

d. $\frac{7\pi}{3}$

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Optimize area of rectangle with perimeter 20: max area $=$

a. 25

b. 20

c. 50

d. 100

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